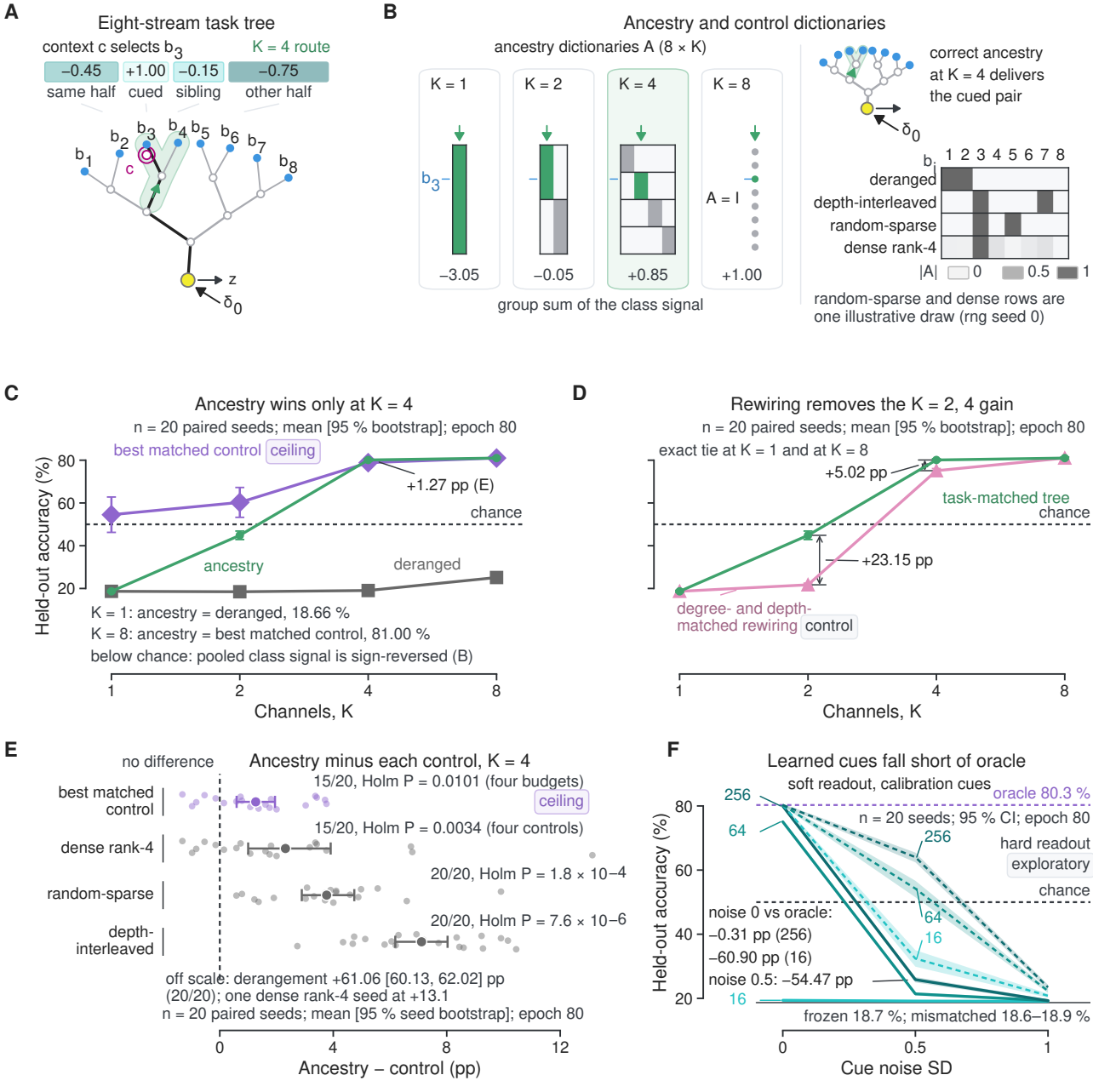

A
Eight-stream task tree
context c selects b3
K = 4 route
−0.45
+1.00
−0.15
−0.75
same half
cued
sibling
other half
b1
b2
b3
b4
b5
b6
b7
b8
c
z
δ0
B
Ancestry and control dictionaries
ancestry dictionaries A (8 × K)
K = 1
K = 2
K = 4
K = 8
b3
A = I
−3.05
−0.05
+0.85
+1.00
group sum of the class signal
correct ancestry
at K = 4 delivers
the cued pair
δ0
b:
1 2 3 4 5 6 7 8
deranged
depth-interleaved
random-sparse
dense rank-4
|A|
0
0.5
1
random-sparse and dense rows are
one illustrative draw (rng seed 0)
C
Ancestry wins only at K = 4
n = 20 paired seeds; mean [95 % bootstrap]; epoch 80
best matched control
ceiling
+1.27 pp (E)
chance
ancestry
deranged
Held-out accuracy (%)
80
60
40
20
K = 1: ancestry = deranged, 18.66 %
K = 8: ancestry = best matched control, 81.00 %
below chance: pooled class signal is sign-reversed (B)
1
2
4
8
Channels, K
D
Rewiring removes the K = 2, 4 gain
n = 20 paired seeds; mean [95 % bootstrap]; epoch 80
exact tie at K = 1 and at K = 8
+5.02 pp
task-matched tree
chance
+23.15 pp
degree- and depth-
matched rewiring control
1
2
4
8
Channels, K
E
no difference
Ancestry minus each control, K = 4
15/20, Holm P = 0.0101 (four budgets)
ceiling
best matched
control
15/20, Holm P = 0.0034 (four controls)
dense rank-4
20/20, Holm P = 1.8 × 10−4
random-sparse
20/20, Holm P = 7.6 × 10−6
depth-
interleaved
off scale: derangement +61.06 [60.13, 62.02] pp
(20/20); one dense rank-4 seed at +13.1
n = 20 paired seeds; mean [95 % seed bootstrap]; epoch 80
0
4
8
12
Ancestry − control (pp)
F
Learned cues fall short of oracle
soft readout, calibration cues
oracle 80.3 %
256
64
n = 20 seeds; 95 % CI; epoch 80
256
hard readout
exploratory
chance
64
16
noise 0 vs oracle:
−0.31 pp (256)
−60.90 pp (16)
noise 0.5: −54.47 pp
16
frozen 18.7 %; mismatched 18.6–18.9 %
Held-out accuracy (%)
80
60
40
20
0
0.5
1
Cue noise SD